Tabular Foundation Models for Data Science Agents

Classification, Regression & Forecasting with TabPFN

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DeepLearning.AI

Same API, different mechanics

scikit-learn compatible — same fit / predict — swap in as a drop-in estimator, no pipeline rewrite

Classical ML

.fit() → Trains a new model — slow, per dataset

.predict() → Applies it — cheap

TabPFN

.fit() → Uploads the training table as context — no training

.predict() → One transformer forward pass over context + new rows

in-context learning — all the compute lives here

Classification: is this patient at risk?

Data about patients

target: disease

age	sex	chest pain	chol	thal	disease
63	Male	typical	233	fixed	0
54	Female	—	286	—	1
41	Male	atypical	—	normal	0
⋮	⋮	⋮	⋮	⋮	⋮
58	Female	non-ang.	—	normal	?

missing value

> No training — the model learns the (historical) table in context

predict()	→ label **1**
predict_proba()	→ **0.81** probability for label **1**

How well did the model do?

	predicted sick	**predicted** healthy
actually sick	Caught true positive	Missed disease false negative the costly one
actually healthy	False alarm false positive one extra test	Correct true negative

Precision of those flagged sick, what's ratio is really sick?

$$\frac{TP}{TP+FP}$$

Recall of the truly sick, what ratio did we catch?

$$\frac{TP}{TP+FN}$$

Every regression is a distribution

sqft	beds	year	price
1,840	3	1998	?

→ TabPFN →

a specific price

345.000

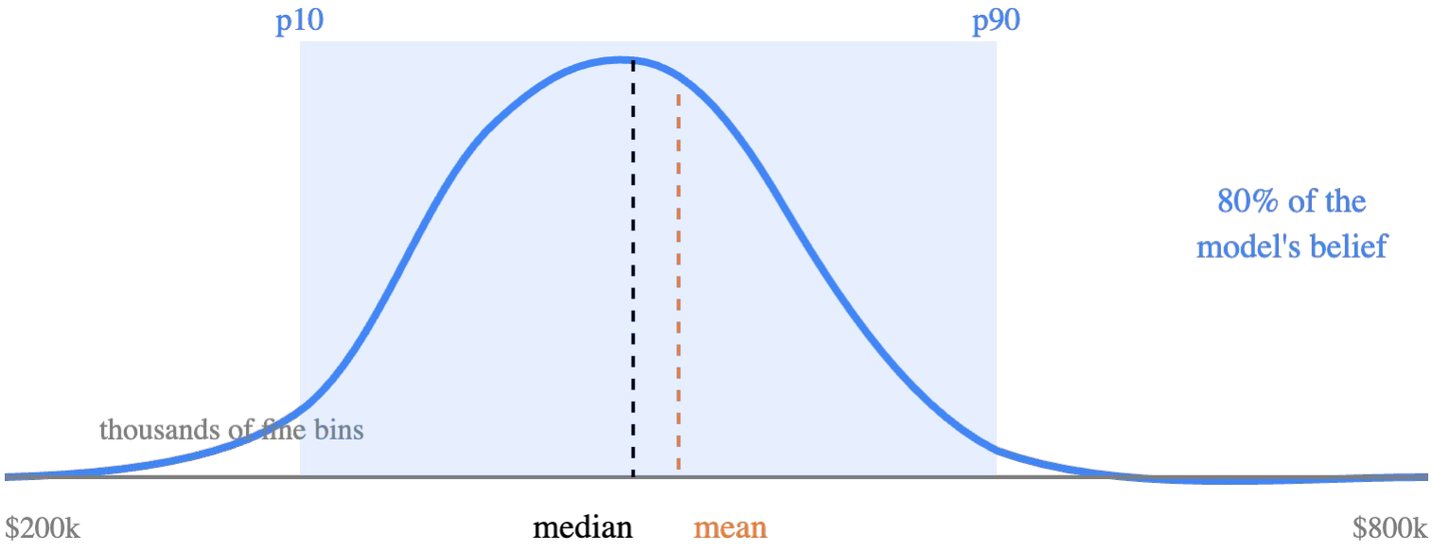
Predict a number: regression



Every regression is a distribution

sqft	beds	year	price
1,840	3	1998	?

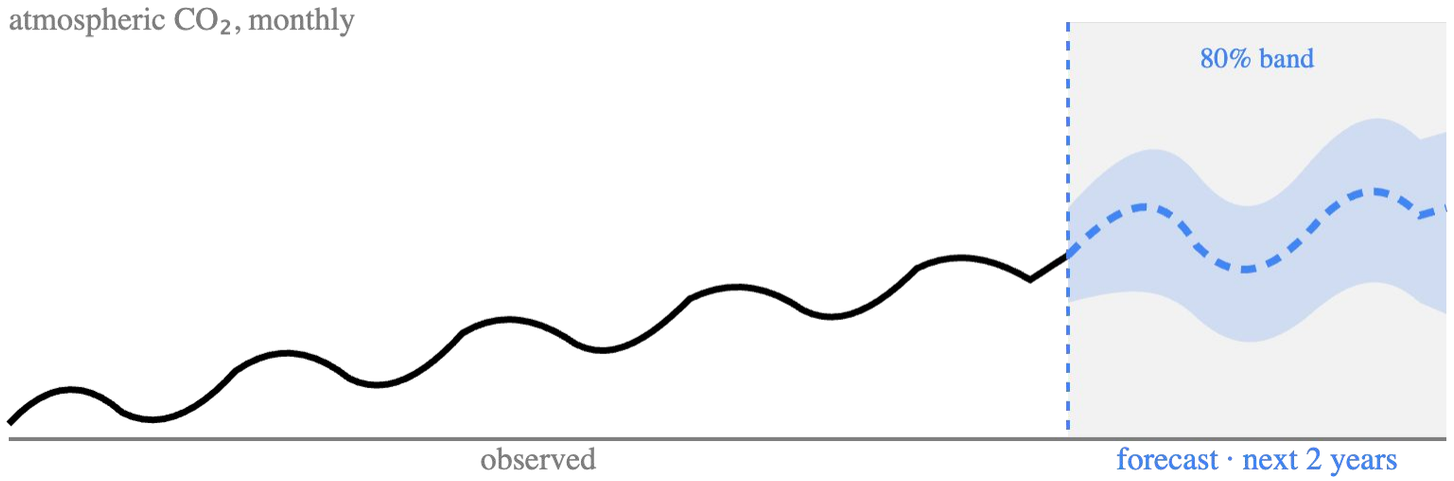
→ TabPFN → a distribution over sale price



> *"My best single guess is the mean — but I'd bet 80% odds the real price lands between the 10th and 90th percentile."*
> *When these two numbers are close to each other, then higher confidence*

A tight band is confidence; a wide band is the model telling you it's guessing.

Time-series forecasting



1 Represent time-series as a tabular problem

2 A forecast is just a table with empty target rows

3 The model only knows what's in its context — adding additional features may help with time-series forecasting